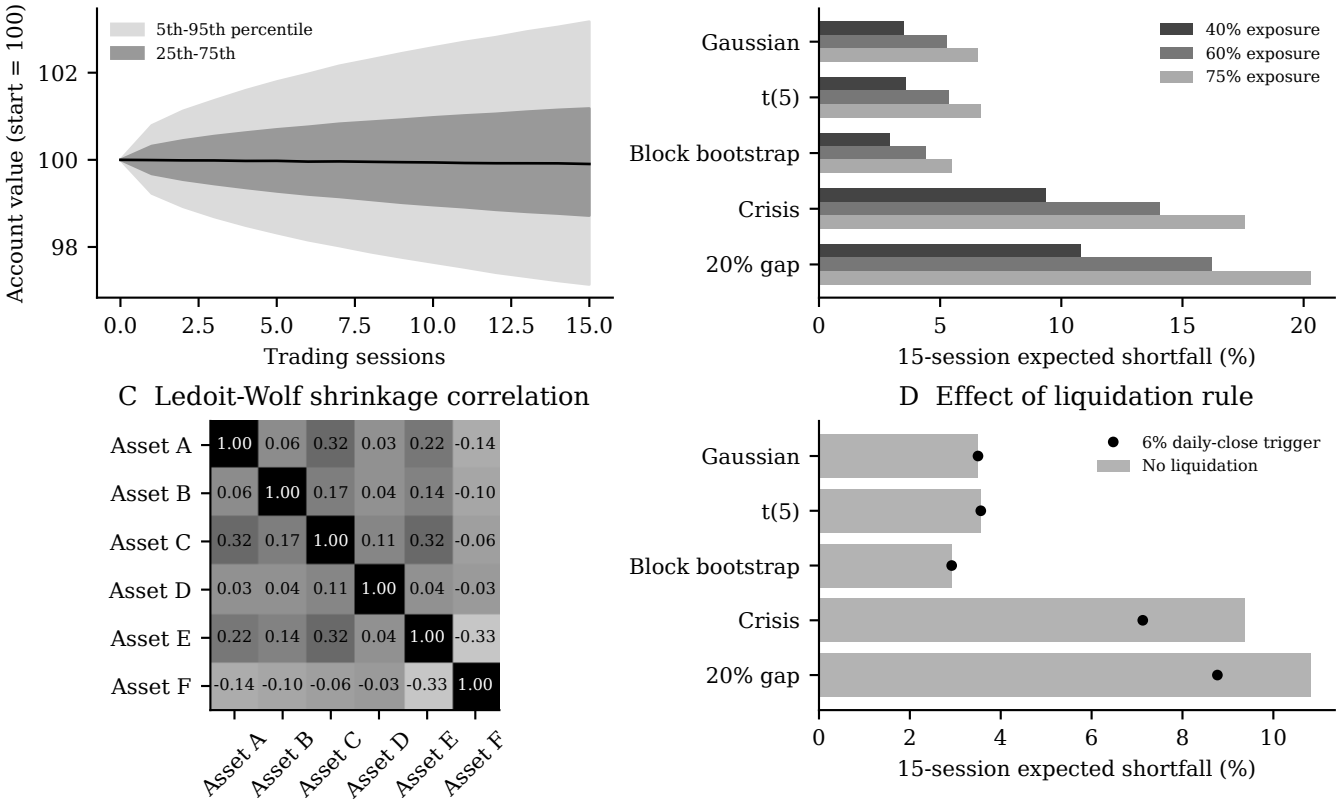
A Calibrated Monte Carlo fan
5th-95th percentile
25th-75th
Account value (start = 100)
102
100
98
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Trading sessions
B Tail-risk model comparison
40% exposure
60% exposure
75% exposure
Gaussian
t(5)
Block bootstrap
Crisis
20% gap
0
5
10
15
20
15-session expected shortfall (%)
C Ledoit-Wolf shrinkage correlation
Asset A 1.00 0.06 0.32 0.03 0.22 -0.14
Asset B 0.06 1.00 0.17 0.04 0.14 -0.10
Asset C 0.32 0.17 1.00 0.11 0.32 -0.06
Asset D 0.03 0.04 0.11 1.00 0.04 -0.03
Asset E 0.22 0.14 0.32 0.04 1.00 -0.33
Asset F -0.14 -0.10 -0.06 -0.03 -0.33 1.00
Asset A
Asset B
Asset C
Asset D
Asset E
Asset F
D Effect of liquidation rule
6% daily-close trigger
No liquidation
Gaussian
t(5)
Block bootstrap
Crisis
20% gap
0
2
4
6
8
10
15-session expected shortfall (%)